Started on	Tuesday, 29 November 2022, 7:22 PM
State	Finished
Completed on	Tuesday, 29 November 2022, 7:25 PM
Time taken	2 mins 57 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

What is the key difference between locks and notifications?

Select one or more:

- a. Unlock must be called by the thread who holds the lock while notifications can be sent from any thread ✔
- b. Notifications are prone to race conditions ✔
- c. Notify must be called by the thread who holds the condition/semaphore lock while unlocks can be sent from any thread

Unlock must be called by the thread who holds the lock while notifications can be sent from any thread

The correct answers are: Unlock must be called by the thread who holds the lock while notifications can be sent from any thread, Notifications are prone to race conditions

Question **2**

Correct

Mark 2.00 out of 2.00

Which of the following synchronization methods uses busy waiting?

Select one:

- a. spinlock ✔
- b. mutex
- c. semaphore
- d. condition variables

The `lock()` operation on a spinlock triggers a busy waiting if the lock is being held by another thread.

The correct answer is: spinlock

Question **3**

Correct

Mark 2.00 out of 2.00

How does the child process(created through `fork()`) differ from its parent?

Select one:

- a. file descriptors
- b. process id ✔
- c. code area
- d. user which created the process

The `fork()` call creates a new process which is an identical copy of the parent process and inherits file descriptors, the memory of the process (including the code) and the user which created the process. It does not inherit the process id.

The correct answer is: process id

Question **4**

Correct

Mark 2.00 out of 2.00

What type of race condition can happen when running the below code snippet in multiple threads?

```
1 if (var < n)
2     var++;
```

Select one:

- a. time of check to time of use ✔
- b. data leak
- c. use after free
- d. dangling pointer

TOCTOU is a type of race condition which involves checking a shared (unprotected) variable value after it has been modified.

The correct answer is: time of check to time of use

Question **5**

Correct

Mark 2.00 out of 2.00

Which of the following represents an advantage of using spinlocks when compared to mutex?

Select one:

- a. Spinlocks use lock/unlock calls with lower time overhead than mutexes. ✔
- b. Spinlocks can be used for larger critical sections.
- c. Spinlocks can be used for multi core systems, while mutexes can not.
- d. Operations from a critical section protected by spinlocks can be blocking.

Spinlocks use the busy waiting mechanism which involves less time overhead than lock/unlock calls made by mutexes.
%~feedback%

The correct answer is: Spinlocks use lock/unlock calls with lower time overhead than mutexes.

◄ Curs 06: Calcul: Recap

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Curs 08: I/O (partea 1) ►